PDF Looker Pro: an open-source annotating PDF reader and document editor

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Abstract

Portable Document Format (PDF) readers are abundant, but open-source readers that *write* into a PDF rather than beside it are not: most either render the page and keep annotations in a sidecar, or depend on a large commercial-grade engine. This paper describes **PDF Looker Pro**, a desktop PDF reader, annotator and page editor whose PDF layer is written from scratch in TypeScript and whose only runtime dependency is the pdf.js rendering engine. We report the engineering rather than a new algorithm, because the interesting failures in this kind of system are invariants that are easy to state and easy to violate. Three are examined in detail: an incremental annotation writer must rebuild a page's `/Annots` from "existing minus replaced, plus new" or it will duplicate every annotation on every save and resurrect every deleted one; a whole-document rewrite must materialise attributes a page previously inherited, or flattening the page tree silently resizes and reorients it; and merging two documents must renumber every reference in the incoming object graph while copying stream bytes verbatim. The annotation model keeps two representations of each mark, a standard subtype for interoperability and a private dictionary for lossless round-tripping, and the paper is explicit about the cost of that duality. The system is verified by 789 behavioural assertions over 31 suites — 734 against the browser build and 55 against the packaged application — and we report the defects that methodology caught, including one the authors introduced into the harness itself, together with the case in which it gave a false sense of coverage. The application is 17,323 lines of source over 47 files, MIT licensed, and publicly available.

Keywords: PDF, document formats, annotation, incremental update, software verification, Electron

1 Introduction

A PDF file is a program for a printer: a page tree, content streams, fonts, and a cross-reference index that says where every object lives. Readers are therefore easy to classify by what they do with those bytes. A *renderer* reads them. A *reader with comments* either writes new objects into the file — which changes the bytes other applications see — or keeps annotations in a companion file, which the reader next to it will never show.

The distinction matters because a PDF is usually not the only copy. An annotated manuscript is mailed to a co-author, a marked-up drawing goes to a fabricator, a reviewed contract goes back to a lawyer. An annotation that lives in a sidecar reaches none of them.

Writing into a PDF is not hard to begin and hard to get right. The format is designed for *incremental* updates: append objects, rewrite the dictionaries that changed, add another cross-reference section pointing back at the previous one, and the file remains valid for every existing reader. The difficulty is that an update is a *diff*, so every save has to know what the previous save did. A writer that only appends is correct exactly once.

This paper reports what it took to build such a reader as a self-contained system: **PDF Looker Pro**, an MIT-licensed desktop application for Windows whose PDF layer — parser, incremental writer, whole-document rewrite and merger — is written from scratch in TypeScript. The only runtime dependency is pdf.js [2], used for rasterisation and text extraction; the page model, the writers and the annotation model are the authors' own. The feature set follows that of a long-standing commercial Windows viewer for which development has ceased [6]; there is no affiliation with it and no code in common.

Contributions. We claim no new algorithm. The contributions are an account of the invariants that a system of this kind must satisfy, the design that satisfies them, and a verification methodology with an honest record of what it caught and what it missed.

1. **A deletion-correct incremental annotation writer** (§4.2). A page's `/Annots` is rebuilt as *(existing* \ *replaced)* ∪ *new*, where "replaced" is the set of object numbers the previous save produced. A page whose annotations were all deleted is still rewritten, because nothing else removes an obsolete entry. We show the two user-visible symptoms of violating this invariant — annotations duplicating on every save, and deleted annotations reappearing — both of which occurred in this codebase and both of which are now covered by a regression test.
2. **A whole-document rewrite that materialises inheritance** (§4.3). Flattening the page tree to support delete, insert, reorder, extract and crop destroys attributes a page inherited from intermediate `/Pages` nodes (`MediaBox`, `CropBox`, `Resources`, `Rotate`, `UserUnit`). The rewrite writes them onto every page, copies objects verbatim when they are stored uncompressed, re-serialises those inside object streams, and recomputes the cross-reference.
3. **Cross-document merging by reference renumbering** (§4.4). The incoming object graph is renumbered above the host's highest object number; every reference is rewritten, every stream's bytes are copied unchanged, and the incoming pages are rebuilt on the host's page tree.
4. **A dual-representation annotation model** (§5.2) that writes a standard subtype so other readers display the mark, and a private dictionary so this reader restores it without loss. We state the cost: two representations can disagree, and the standard one cannot express everything the private one can.
5. **Search over a virtualised document** (§6) that highlights by rewriting the text layer's spans, so a painted match is aligned with the painted glyphs by construction, guarded by a comparison against the strings the index was built from.
6. **A dual-target verification methodology** (§7): 789 behavioural assertions over 31 suites, 734 driving a browser build and 55 driving the packaged Electron application as a real executable, with a case study of a defect the second target caught and of three defects the authors introduced into the harness itself.

Scope. The paper describes a desktop application, not a library or a format extension. It does not claim conformance to any profile of ISO 32000 [1], does not handle encrypted documents for writing, and does not implement forms, digital signatures or optical character recognition. Section 9.2 lists the limitations in full; we regard that list as a deliverable, not a disclaimer.

Availability. Source, tests, build scripts and the measurement scripts that produced every number in this paper are at `https://github.com/dongfangxy/pdf-looker-pro`, MIT licensed. The packaged Windows build is published as a release artefact of the same repository.

2 Background and related work

The format. PDF is specified by ISO 32000-1 [1]. A file is a sequence of indirect objects, a page tree under a catalogue, and a cross-reference table that maps object numbers to byte offsets. Two properties

Table 1: Dependencies. The application's only runtime dependency is the rendering engine. Everything else is a build or verification tool and ships in no artefact.

Package	Role	Version	Licence
`pdfjs-dist`	rendering and text extraction	5.x	Apache-2.0
build and verification only			
`electron`	desktop host	44.x	MIT
`electron-builder`	packaging	26.x	MIT
`vite`	bundler	7.x	MIT
`typescript`	type checker (erased at build)	5.9	Apache-2.0
`playwright`	browser and Electron driving	1.64	Apache-2.0

matter here. First, the format admits *incremental updates*: a new section may be appended at the end of the file with its own cross-reference table and a `/Prev` pointer to the previous one, so the original bytes never move. Second, page attributes are *inheritable* along the page tree: a page that omits `/MediaBox` takes the nearest ancestor's.

Streams are stored as a dictionary followed by raw bytes whose length the dictionary declares. The bytes may be compressed by any of several filters, and objects may be packed into a compressed container called an object stream. A writer that wants to copy a stream without decoding it must therefore preserve the bytes exactly and copy the filter description; a writer that has to rewrite a reference *inside* an object stream cannot do so without decoding and re-encoding the container.

Rendering engines. pdf.js [2] is a JavaScript PDF renderer, Apache-2.0 licensed, and the rendering engine of Firefox. It exposes a page model and a canvas rasteriser, plus a text-extraction API that yields positioned glyph runs. MuPDF [3] is a C engine, dual-licensed AGPL/commercial, with annotation support. PDFium is the engine inside Chromium. These are renderers first; writing is either absent or a secondary concern.

Readers and annotators. Okular [4] (KDE, Poppler-backed) and Evince (GNOME) are mature open-source readers with annotation support that write annotations into the document through their respective back ends. Apache PDFBox [5] and iText are libraries with full writing support, used by other applications rather than being applications themselves. PDF-XChange Viewer [6], whose feature set this project follows, is a commercial Windows viewer that is no longer developed.

The gap this project occupies is deliberately narrow: an *application*, in which the reader, the writers and the annotation model are open and readable together, with no engine dependency beyond rasterisation, and with the writing logic documented as a set of invariants rather than as behaviour to be discovered. We do not claim to be better at rendering, and we do not compare rendering quality against the systems above; §8 explains why such a comparison would be neither fair nor useful at this stage.

Why write at all. A sidecar annotation file is simpler in every engineering respect: no incremental updates, no object numbering, no risk of corrupting the document. It is also invisible to every other application. The design decision recorded here is that the document is the artefact that travels, so the annotations must travel inside it. The price is paid in §4.

3 System architecture

The application is an Electron [7] shell around a TypeScript renderer. The shell owns windows, the native menu, file dialogues and printing; the renderer owns everything else. The renderer is bundled by Vite [8] into a single script of 660.2 kB (197.8 kB gzip) plus 26.1 kB of CSS (6.0 kB gzip). User-interface

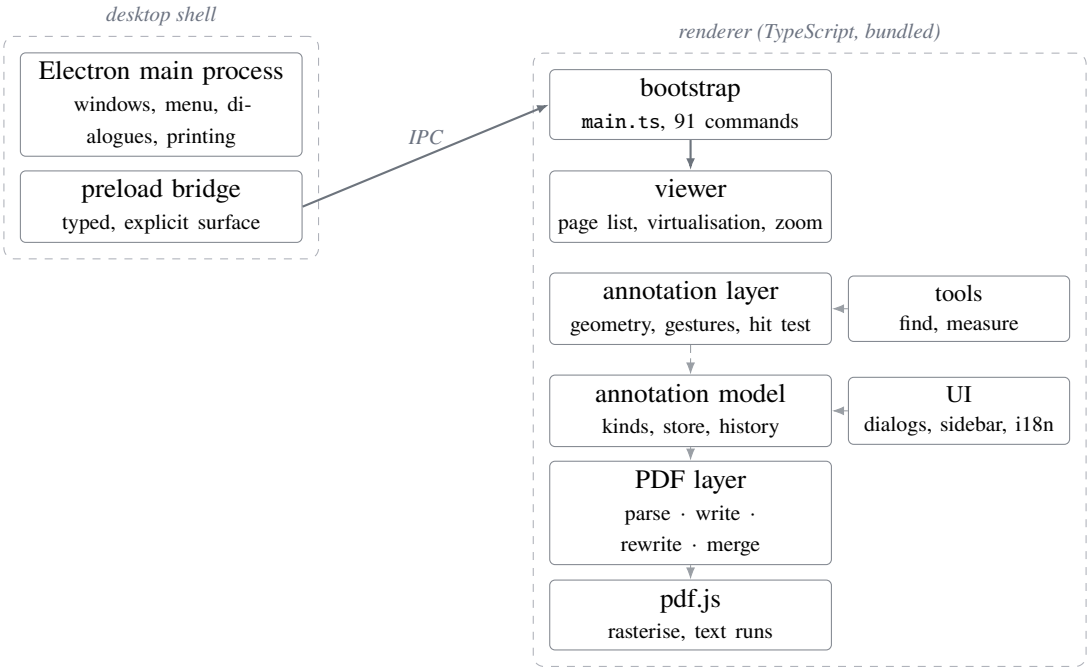


Figure 1: Module layout. The PDF layer is the only component that touches bytes, and it is independent of the renderer: it parses into its own value types rather than using pdf.js's object model, which is what allows the writers to copy objects the renderer never looks at. Solid arrows are calls; dashed arrows are use.

resources — icons and strings that are not code — are packed into a 19 kB container with a format of the project's own, described in the repository.

Figure 1 shows the layout and Table 2 its size. Two decisions shape everything else.

The PDF layer has its own object model. pdf.js exposes a document proxy that can resolve objects, but it does not expose byte offsets, object generations, or the raw text of an uncompressed object. All three are needed to write an incremental update: the offset is what a cross-reference entry holds, the generation is part of a reference, and copying an uncompressed object's bytes verbatim is both faster and safer than re-serialising a parse of it. The project therefore has its own lexer and parser (`src/pdf`, eight files), and uses pdf.js strictly as a rasteriser and text extractor. The duplication is real — two parsers over the same bytes — and is the largest structural cost in the system; it buys independence from a renderer's internal model, which is the property the writers need.

A page-space coordinate system of our own. The renderer, the annotation model and the hit test all work in *unrotated page space at scale 1, origin top-left, y down*. PDF user space has its origin at the bottom-left and is scaled by the page's `UserUnit`, so every conversion is explicit ($y_{\text{pdf}} = \texttt{MediaBox}_3 - y$) and every annotation stores coordinates that survive a zoom, a rotation or a change of view mode without being rewritten. This sounds pedantic and is not: the first implementation mixed rotated and unrotated coordinates and placed marks correctly at 100 % and nowhere else.

The 1,414 lines of CSS deserve a note: the application has four themes (light, dark, sepia, high contrast) driven by custom properties on a `data-theme` attribute, which is why the stylesheet is larger than several TypeScript modules. The 3,887-line bootstrap is the concentration of failure in this architecture — 91 commands, their bindings, the menu descriptor, the debug hooks and the wiring of every module. It is also the file most likely to be touched by a change, and the one where an omission is silent: a command registered without a control appears in no menu and reports nothing.

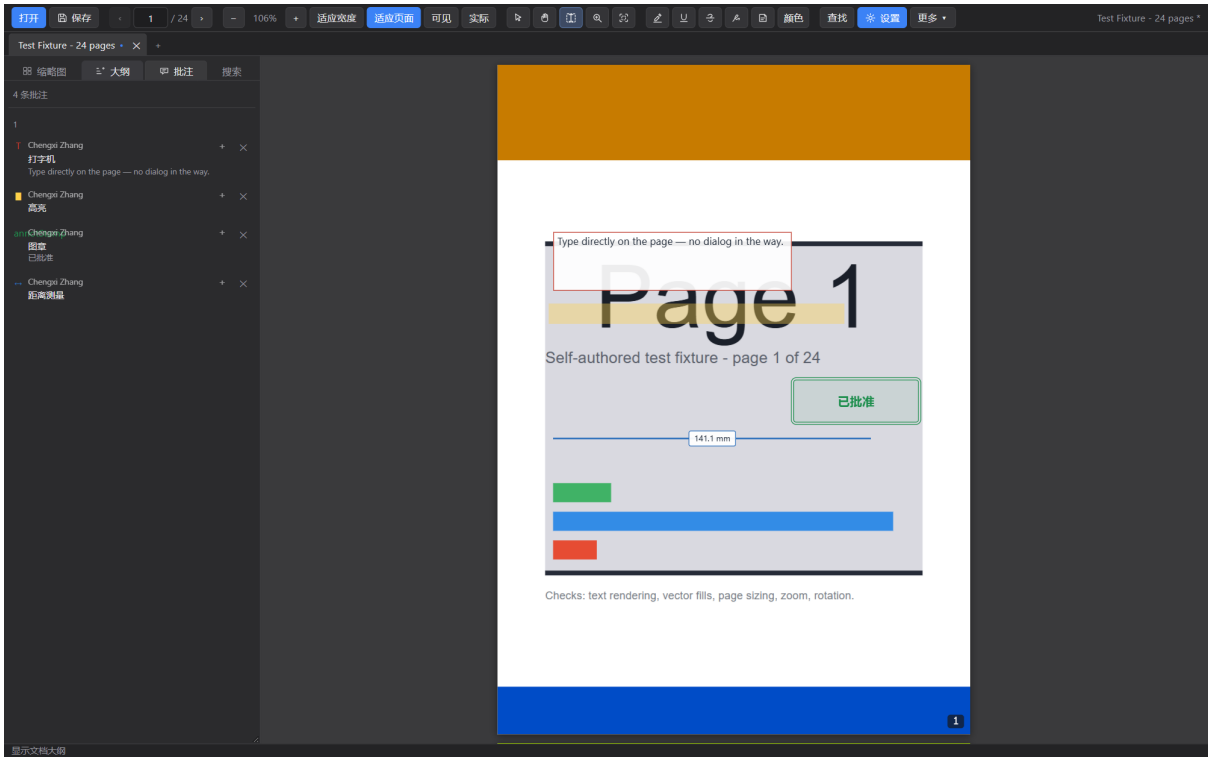


Figure 2: The application. The page carries a typewriter box, a text highlight, a standard stamp and a distance measurement; the sidebar lists them by kind, page and author. The panel matters as much as the page: an annotation that cannot be found again is not much of an annotation.

4 The PDF layer

Five operations touch bytes: parse, incremental write (annotations), whole-document rewrite (page operations), merge, and flatten (annotations baked into page content). They share a parser and differ in what they are allowed to assume.

4.1 Parsing

The parser (`src/pdf/syntax.ts`, `document.ts`) handles the parts of ISO 32000 that real documents use: classic cross-reference tables, cross-reference streams, object streams, incremental update chains via `/Prev`, and the standard filters for the streams it must decode (none are decoded for a verbatim copy). It retains for every object both the parsed value and, when the object was stored uncompressed, the exact source text.

The parsed value distinguishes streams by carrying their raw bytes:

```
type PdfStreamValue = {
  kind: 'stream';
  dict: PdfDict;
  raw: Uint8Array;   // exactly as it appears in the file, before any filter
};
```

This one field is what makes merging possible without a codec: a stream from another document can be written back with its bytes untouched, because the only thing that needed rewriting was the dictionary's references. See §4.4.

Two parser details caused defects worth recording, because both are invisible until a file is written and re-read. First, an end-of-stream marker has two positions — where the `endstream` keyword begins and where its preceding newline ends — and slicing content with the trimmed one left a trailing newline that looked like the start of the next token. Second, object numbers are not necessarily contiguous: a file

Table 2: Source size. Lines are physical lines of `.ts`, `.mjs`, `.cjs` and `.css`; blank lines and comments are included, so the figures overstate logical code by roughly a fifth. Measured over the 47 application files and the 52 test and tooling files.

Component	Files	Lines
PDF reader and writers (`src/pdf`)	8	3,349
Annotations (`src/annotations`)	5	2,146
Viewer (`src/viewer`)	5	1,334
UI, dialogs (`src/ui`)	5	1,232
Tools: find, measure, snapshot (`src/tools`)	5	972
i18n and command table (`src/i18n`)	7	1,382
Sidebar panels (`src/sidebar`)	4	738
Tabs and sessions (`src/tabs`)	2	178
Saved searches (`src/search`)	1	91
Styles	1	1,414
Bootstrap (`main.ts`, `product.ts`)	2	3,887
Electron host (`main.cjs`, `preload.cjs`)	2	600
Application	**47**	**17,323**
Tests and tooling (`tools/`)	52	10,594

may reference an object it does not define, and the parser must distinguish "missing" from "null", since a page's `/Annots` may legitimately be absent.

4.2 The incremental annotation writer

Saving annotations appends objects to the existing file, rewrites the page dictionaries that gained or lost annotations, and appends a cross-reference section whose trailer carries `/Prev` pointing at the previous one. The original bytes are never modified, so the operation is reversible by truncation and a reader that ignores the new section still sees a valid document.

The invariant that governs everything is how a page's annotation array is computed. Let E be the object numbers currently listed in the page's `/Annots`, R the object numbers this writer produced for that page on the previous save, and N the object numbers of the annotations now in the model. Then

```
annots = [ refs(E) \ refs(R) ] ++ [ refs(N) ]
```

and the page dictionary is rewritten whenever $N \cup R \neq \emptyset$ — including when $N = \emptyset$, because a page whose annotations were all deleted has no other reason to be rewritten, and its `/Annots` would otherwise keep pointing at them.

What goes wrong without it. The first implementation appended: `annots` $= E \cup N$. Two symptoms followed, both reported by a user before any test noticed.

1. **Duplication.** Every save appended a fresh copy of every annotation on the page and listed both. Because the copies are byte-identical and coincide geometrically, the page looks correct; the file grows, and the reader may select one copy while deleting another. This reads to a user as "the delete button does nothing".
2. **Resurrection.** A deleted annotation was removed from the model but its object was still listed in E, so the next save re-published it. The mark could not be removed from the document by any sequence of operations in the application.

The fix records, per page, the object numbers the writer produced, keeps that map on the document session, and passes it back on the next save as R; the writer uses the numbers it produced this time as the

new *R*. Objects superseded by earlier saves remain in the file as orphans — unreferenced, legal, and the price of never rewriting the original bytes. §7 describes the regression test, and a note on the test itself: the first version of that test counted annotations in the file by calling the writer, which is a write, so the counting added copies. It now fetches the saved file and resolves the page's `/Annots`, counting what the file points at rather than what it contains.

Interoperability versus fidelity. Every annotation is written twice, in a sense: once as a standard annotation object that other readers display, and once as a private `/PLP` dictionary holding this application's exact kind, stroke width, dash pattern and font size. Standard subtypes cannot express all of it — a text box and a callout are both `/FreeText`, a replaced word is a `/StrikeOut` — so the private entry is what makes a round trip lossless. Section 5.2 discusses the cost of this duality.

4.3 Whole-document rewrite

Delete, insert, reorder, extract, crop and rotate pages cannot be expressed as an appending update if they move pages between documents or change the page order: the page tree has to be rebuilt. `rewritePdf` takes a plan — for each output page, a source page index plus optional rotation, crop box and a flatten step — and produces a new file.

The invariant here is about *inheritance*. In the source, a page may take `MediaBox`, `CropBox`, `Resources`, `Rotate` and `UserUnit` from an intermediate `/Pages` node. The rewrite produces a flat page tree with one `/Pages` node, so each page must carry everything it inherited. Omitting this does not fail: the file parses, and the pages are the wrong size, or sideways, which is a far worse failure mode than an error.

Object handling follows from what the parser retained:

- an object stored uncompressed is copied as its exact source text, with no re-serialisation, so anything the application does not model survives;
- an object inside an object stream is re-serialised from its parsed value, because its original bytes are inside a compressed container;
- a source page object is reused, and its dictionary rewritten, when that page appears exactly once in the plan; otherwise a fresh object is written;
- pages removed by the plan leave their objects unreferenced — again legal orphans.

Because the plan can reference the same source page more than once, a rotation of a single page is a one-page plan and a reorder is an identity plan with a permuted index list; the same code path serves all of them.

4.4 Merging two documents

Merging is the one operation that copies between two object graphs. Every reference inside the incoming document must be renumbered to its new home, or it will point at whatever the host happens to keep at that number.

The algorithm is: build a map from each incoming object number to a fresh number above the host's highest; for every incoming object except the page objects, its `/Pages` node and its catalogue, re-serialise the value through a recursive reference rewrite, keeping stream bytes and correcting `/Length` to the byte count; then build the incoming pages on the host's page tree, with the same materialisation as §4.3.

Three details are easy to get wrong and were:

1. **Stream bytes are not references.** A stream's data may contain any bytes, including ones that look like `N 0 R`. Only the dictionary is rewritten.
2. **The incoming page tree is discarded, not merged.** Keeping the incoming `/Pages` node would leave a second tree with no parent, and readers differ on what to do with it.

3. **Insertion is at a position, not an append.** Merging a cover sheet is the common case, so the interface takes a page index and clamps it to the document rather than rejecting it.

4.5 Flattening and summary documents

Flattening bakes annotations into page content: an extra content stream is appended rather than merged into the existing one (a stream's bytes are immutable, and concatenating inside one is a rewrite of bytes the writer promised not to touch), the page's `/Annots` is dropped, and a Helvetica font is installed into a *copy* of the page's resource dictionary so the page's own font numbering is not disturbed.

Summarising produces a new document containing the annotated pages. The obvious implementation is a text listing of the comments, and it fails on the first document whose comments are not in a Latin script: a text summary needs a font that covers whatever the reader typed, and embedding one is a large amount of machinery for a summary. The implementation instead renders each annotated page — with its annotations already flattened in — at twice the page size and assembles the images into a new PDF. The result carries the marks and the words inside them in any language, needs no font of its own, and costs nothing that the printing path did not already provide.

5 The annotation model

5.1 Geometry

An annotation stores its geometry in unrotated page space at scale 1, origin top-left, y increasing downwards. Four consequences follow, all of them the reason for the choice:

- a zoom changes a scale factor and nothing else;
- a rotation changes the transform used to paint, not the coordinates stored;
- a view-mode change (single, continuous, facing, facing-continuous) does not touch the model at all;
- hit testing compares a point in the same space, so it is a rectangle test rather than an inverse projection.

PDF user space is the mirror image, with an origin at the bottom-left and a scale given by the page's `/UserUnit`, so writing an annotation is an explicit `toScreen`/`fromScreen` conversion in one place (`src/annotations/layer.ts`). Keeping that conversion in one function, rather than inline at each use, is what made it possible to add rotated pages without auditing a dozen call sites.

5.2 Kinds and the dual representation

Fifteen kinds are modelled. Twelve of them draw or carry text: highlight, underline, strikeout, *replace text*, freehand ink, sticky note, rectangle, ellipse, arrow, typewriter (the original product's name for a free text box), callout text, and stamp — where stamp takes one of six standard names. The remaining three are measurements: distance, perimeter and area.

Each is written as a standard subtype, so that other readers display it: `/Highlight`, `/Underline`, `/StrikeOut`, `/Ink`, `/Text`, `/Square`, `/Circle`, `/Line`, `/FreeText`, `/PolyLine`, `/Polygon`, `/Stamp`. Alongside, a private `/PLP` dictionary carries `/K` (the exact kind), `/W` (stroke width), `/D` (dash) and `/F` (font size). Readers that do not know `/PLP` ignore it; this one prefers it on load and falls back to inferring a kind from the subtype, which is how annotations made in other applications are displayed with sensible defaults.

The cost of two representations. This duality is the model's weakest point and worth stating plainly. A third-party `/StrikeOut` with no `/PLP` becomes a strikeout in this reader, which is correct; a `/StrikeOut` written by this reader for a *replace text* annotation becomes a strikethrough everywhere else, which is the closest the format can express and is what the paper's §4 describes as a limitation rather than a feature. Two representations can also disagree — a file edited in another application may carry a `/PLP` kind that

its subtype no longer supports — and the loader resolves that in favour of the subtype when the two are incompatible, silently. We know of no such file; the rule exists because guessing in the other direction would be worse.

Threads. A reply is its own annotation, a `/Text` object whose `/IRT` points at its parent through an indirect reference; this is the mechanism every reader uses to display a conversation. Reading back therefore takes two passes: a reply may reference a parent that appears later in the page's list, so the loader collects replies first and attaches them once every annotation is known. A reply whose parent lives on another page is still shown, rather than dropped, because a missing parent is a fact about the file rather than about the reply.

Identity and deletion. Annotations carry a locally generated identifier and, separately, the object number they were read from. The two must not be conflated: the identifier survives an edit, while the object number is what a save must remove. Keeping the second only on the document session, not on the annotation, is the design that made §4.2 correct.

5.3 Interaction

Gestures are owned by the annotation layer rather than by the tool manager, because they differ per kind: text markup consumes a selection, ink and rectangles drag out geometry, a note is a single click, and a stamp is a click with a size of its own.

Two interaction rules were learned from defects:

- **A click selects an annotation under any tool**, and the *gesture* belongs to the active tool. An earlier version stopped propagation whenever a pointer landed on an annotation, which broke drag-to-scroll and marquee selection for gestures that merely started on top of a mark.
- **A text tool pressed on an existing text box edits it** rather than stacking a blank box on top. The earlier behaviour selected the annotation and then covered it with an empty box, which a user reasonably reported as "I cannot select my annotation".

5.4 Rendering and printing

Pages are virtualised: only pages near the viewport have a rasterised canvas, and a page leaving the window drops its canvas while keeping its element. §8 shows the consequence in the measurements — opening a 200-page document and a 24-page document take the same time to first painted page.

Printing rasterises the selected range at twice the page size (144 DPI) into a hidden container, hides the application shell with a print stylesheet, and calls the platform's print dialog; in the desktop build the same is done through the Electron main process. The printed sheets are removed when the print job reports completion rather than on a timer, because a timer loses the race on a slow printer.

6 Search

6.1 Matching and highlighting

The document's text is not available as a string; it is a set of positioned glyph runs in a text layer. The find index concatenates each page's run strings into one haystack and records the run boundaries, so a match is an offset into a page's concatenated text plus a length.

Highlighting does not compute its own geometry. It rewrites the text layer's spans, splitting a span at the match boundaries and wrapping the remainder in a `<mark>`-like element, which is the technique pdf.js uses internally. The benefit is that a painted match is aligned with the painted glyphs *by construction*: there is no second coordinate computation that could disagree with the first.

Because the index and the layer are separate, they can disagree. Before painting, the page's live span strings are compared with the strings the index was built from; a page that does not match is skipped rather than mis-highlighted. In practice this triggers when a page's text layer is re-rendered between indexing and searching — on a fast scroll through a document whose index is still being built — and skipping is the correct answer, since the offsets no longer mean what they did.

6.2 Result presentation

Results are grouped by page, in document order, each row quoting the words around the match. The quote comes from the index itself rather than from re-extracting the page: the offsets a hit holds are offsets into exactly that string, so the snippet cannot drift from the highlight.

Two ordering defects are worth recording because both produced a screen that contradicted itself. First, the panel was told about a query *before* the search ran, so it redrew from the previous query's matches; because the zero-hit path returns early, nothing redrew afterwards, and a search for nonsense left twenty-two results beside a status bar saying there were none. Second, a match list is a view of mutable state: the "current" row must be re-derived whenever the index changes, not only when the user moves to the next match.

6.3 Folder search across documents

Searching a folder means reading files the user has not opened. The application's desktop shell refuses to read any path the user has not granted, so listing a folder also *grants* the files it finds: a folder the user types is as explicit a permission as one chosen from a dialog, and the grant is what lets the scan proceed. Depth and file count are bounded, and a file that will not parse is skipped rather than fatal — one damaged document in a folder should not stop the search.

Each file is opened on its own and released again. A folder search is a scan, not a shelf of open documents; holding every parsed document would cost far more memory than it saves, and §8 shows the index is cheap enough to rebuild.

6.4 Saved searches

A saved search stores the question, not the answer: the query, the case and whole-word flags, and the folder when there was one. The hits go stale the moment a document changes, and re-running is the point. Saving under a name that already exists replaces it, so re-saving updates rather than accumulating near-duplicates, and an entry with no name or no query is dropped on read rather than repaired.

6.5 Form fields, without a form layer

The viewer highlights form fields — tinting the rectangles of `/Widget` annotations so a reader can see where a form asks for input — and does not claim to fill them. The distinction matters for the paper because it is the honest boundary of the feature: detecting a widget needs the page's annotation array, while filling one needs the field's own hierarchy (an `/AcroForm` dictionary, field types, appearance streams, calculation order and validation), which this application does not implement. Highlighting without filling is useful and truthful; a half-implemented form layer would be neither.

7 Verification

7.1 What is asserted

The project has no unit-test pyramid. It has **31 suites of behavioural assertions**, each of which drives the real application — a real browser for the renderer, the real packaged executable for the desktop build — and then asks questions of what it produced: pixels, bytes, and document model state.

Table 3: The verification suites, grouped. The per-suite breakdown is in the repository (`paper/metrics/results.json`, produced by `tools/collect-metrics.mjs`).

Group	Suites	Assertions
Foundations: resource container, embedded font, i18n packs	3	78
Reading and interface: layout, view modes, zoom, print, themes, tabs	9	297
PDF writing: incremental save, round trip, rewrite, page operations, merge	8	202
Annotations: author, stamp, reply, replace, flatten, summary, fields	6	116
Search: in-document, saved	2	41
Browser build	28	734
Packaged application	3	55
Total	**31**	**789**

The choice follows from where the defects were. In a system that parses, rewrites and re-parses a binary format, the interesting failure is rarely a wrong return value; it is a file that opens and is wrong. Two examples from the suites:

- After a save and a reopen, a text annotation is clicked with a real mouse and the assertion is that the annotation is selected and shows a resize grip — not that a function returned an object.
- After flattening, the styled text is sampled from the rendered canvas at the coordinates the annotation occupied, and the sampled colour must be the annotation's colour. A flattening bug that wrote the operators in the wrong coordinate system produces a valid file and fails this assertion.

Snapshots are deliberately not used: they would freeze rendering differences between browser versions and say nothing about whether a saved file is correct.

7.2 Two targets

The renderer is verified in a headless browser; the desktop shell is verified by launching the packaged executable through Playwright's Electron support. Both are necessary because they fail differently. The browser build exercises the code paths a web deployment would take, and the packaged build is the only place where file dialogues, native menus, window geometry memory and the print path through the main process exist at all.

The split, measured, is 734 assertions over 28 suites for the browser build and 55 over 3 suites for the packaged application; Table 3 groups them.

7.3 Defects the methodology caught

The one it did not catch first. The annotation duplication and resurrection described in §4.2 was reported by a user after release, not found by the suites. The reason is instructive: the round-trip suite asserted that annotations survived a save and reopen — which they did, twice over. Nothing asserted that saving *twice* changed nothing, and nothing asserted that a deleted annotation was gone from the file. Both assertions now exist, in `e2e-save-reopen`: save two, reopen, click one, save twice more, delete one, save, reopen, delete the last, save, reopen.

Defects in the harness itself. Three failures were in the measurement rather than the product, and all three are worth recording because each produced a confident wrong answer.

1. **Counting by writing.** The first version of the regression test counted the annotations in the saved file by calling the writer. The writer appends, so the act of counting added a copy, and the test failed for a reason of its own making. The test now fetches the saved bytes and resolves the page's `/Annots`, which counts what the file *points at* rather than what it contains.

2. **Testing the projection.** Two features — an annotation author and a stamp name — were briefly diagnosed as broken because the test read state through a hand-written debug projection that listed a fixed set of fields, and the projection had not been extended. The feature worked; the projection lied. This is the most expensive mistake in the project's history, because it is invisible: a projection tested through itself proves only that the projection works. The rule now is that assertions read the document model directly wherever a hook is not the thing under test.
3. **An unwritten fixture.** Continuous integration went red for three consecutive commits because the fixtures a suite opens are generated by a script, the fixture directory is intentionally not committed, and a new generator had not been added to that script. The suite timed out waiting for a page that was never served. The CI log named the suite; nothing named the cause.

The limits of the harness. Two are known and reported rather than papered over. Synthetic mouse-button events do not reach the packaged window through the automation layer, so the packaged suite asserts what the harness can prove (the tool is selectable, the region exists) and does not claim to have dragged anything. And the window-geometry assertion is stateful: it fails when three suites run back to back with the process killed between them and passes when run alone, because it reads geometry a previous run wrote.

7.4 Reproducing the numbers

Every figure in this paper comes from a script in the repository, not from a notebook. `tools/collect-metrics.mjs` runs the 31 suites and parses each one's summary line into `paper/metrics/results.json`; `tools/benchmark.mjs` produces the timings in Table 5 and writes `paper/metrics/benchmark.json`. Both refuse to guess: a suite that fails is recorded as failing and a timing that cannot be taken is `null`.

8 Evaluation

8.1 What is measured, and what is not

Three things are measured: the size of the system, the size of what ships, and the wall-clock cost of the operations the paper describes. All three are taken from the artefact on one machine (Windows 11, Node 24, an x86-64 desktop CPU) with the scripts named in §7.

We do **not** report a comparison against other readers, and the reason is not modesty. A fair comparison would need a matched document corpus, a defined set of tasks, and a measurement of correctness as well as speed — and this system has no claim to be faster than an engine written in C. What the numbers below establish is narrower and, we think, more useful: that the writing operations are cheap enough to run on every save, and that virtualisation makes open time independent of document length.

8.2 Size

Table 4 separates two facts that are usually conflated. The code this paper describes is 198 kB compressed; the download is 108 MB, and 99.8 % of it is the Electron runtime. That ratio is a property of the packaging choice, not of the application, and it is the project's largest engineering compromise: a native shell would ship a few megabytes and cost months of platform work.

The test-to-application line ratio is 0.61:1. We report it because it is a design choice with a cost — a tenth of the codebase exists to check the other nine tenths — and because a figure that low is only defensible if the suites assert behaviour rather than implementation, which §7 argues they do.

8.3 Cost of the writing operations

Two rows carry the paper's practical argument.

Table 4: What ships. The unpacked executable is the application plus the Electron runtime; the renderer bundle is the code this paper describes.

Artefact	Size
Renderer bundle, JavaScript	660.2 kB (197.8 kB gzip)
Renderer bundle, CSS	26.1 kB (6.0 kB gzip)
UI resource container	19.0 kB
Application source	17,323 lines over 47 files
Test and tooling source	10,594 lines over 52 files
Portable executable (single file)	108.1 MB
Unpacked executable	235.6 MB

Table 5: Measured cost of the operations this paper describes, in the renderer, excluding file I/O. Median of repeated runs on fixtures of 24 and 200 pages; the raw samples are in `paper/metrics/benchmark.json`.

Operation	Document	Median
Open to first painted page	24 pages	271 ms
Open to first painted page	200 pages	279 ms
Build full-text index	24 pages	4.9 ms
Build full-text index	200 pages	40.9 ms
Incremental annotation save	24 pages	1 ms
Whole-document rewrite	24 pages	2 ms
Merge two 24-page documents	48 pages out	3 ms
JavaScript heap after opening	200 pages	7 MiB used

Open time is independent of length. 271 ms for 24 pages and 279 ms for 200 pages — a 3 % difference across an eight-fold increase in document size. This is what virtualisation is for: the application rasterises the pages near the viewport and nothing else, so the cost of opening is the cost of the first screen plus parsing the cross-reference, not the cost of the document.

Writing is cheap enough to do on every save. An incremental save of a 24-page document is around a millisecond; a full rewrite of the same document is two; merging two 24-page documents into 48 is three. These are single-digit milliseconds for operations that earlier sections described as delicate, which is the useful observation about the design: the difficulty was never performance, and the invariants in §4 cost nothing to uphold. A save can therefore be conservative — rewrite the page's annotation array, drop the objects it replaces, write a fresh cross-reference — rather than clever.

The index figure scales linearly: 4.9 ms for 24 pages, 40.9 ms for 200, i.e. about 0.2 ms per page. That is why folder search rebuilds an index per file rather than storing one (§6.3), and why the 7 MiB heap figure for a 200-page document is consistent with holding no more than the pages near the viewport.

8.4 Threats to validity

The measurements are from one machine and one rendering engine, and the electron and browser versions are pinned only by `package.json` ranges. The timings are in-process and exclude disk and IPC, so they understate what a user waits for on a save; the open-time figures include navigation and first paint, so they overstate what a second open of the same file costs.

The assertion count is a measure of effort, not of coverage. There is no instrumentation-based coverage figure in this paper because none was collected; a suite count can be raised by writing easier tests, and the reader should treat 789 as evidence that many behaviours are checked, not as a percentage of the behaviours that exist.

9 Discussion, limitations and conclusion

9.1 What generalises

Three lessons seem portable beyond this application.

An incremental format turns every writer into a stateful operation. The bug in §4.2 is not a coding error in the ordinary sense; it is a failure to model the writer as having a *past*. A function that appends is correct once, so any provider of such a function must answer the question "what did you write last time?" The answer in this system is a per-page set of object numbers held on the document session, and the general statement is that the state belongs next to the bytes, not inside the writer.

Flattening a tree materialises inheritance. This is true of any format with inheritable attributes — CSS, DOM, scene graphs, PDF page trees. The failure is silent because the flattened document is still well-formed; the flaw is that it means something else.

A test can be wrong in a way that looks like a finding. The projection trap in §7 cost the project more time than any product defect, twice, on the same feature pattern. The rule we now apply — assert against the model, not against a debug surface that describes the model — is cheap and should have been adopted on the first occurrence.

9.2 Limitations

This list is the boundary of what the paper's claims cover.

- **No installer and no code signing.** The application ships as a portable executable; Windows SmartScreen warns on first run. Fixing this needs a code-signing certificate, not code.
- **Windows only.** The packaged target is Windows x64. Nothing in the renderer is platform-specific, but the shell is not exercised elsewhere.
- **Encrypted documents are refused for writing.** They open for reading through the renderer; an annotation save against an encrypted file is rejected rather than attempted.
- **Files grow with every save.** The incremental writer never rewrites the original bytes, so superseded objects remain as orphans. A compaction pass would need a whole-document rewrite and is not implemented.
- **Flattening cannot encode text outside Latin-1.** The embedded appearance font is a standard Helvetica; a flatten of Chinese or Greek annotation text renders as placeholders. The summary feature avoids this by rendering pages instead of writing text (§4).
- **No form filling, digital signatures or OCR.** Form fields are highlighted, and §6.5 is explicit about why that is not the same as supporting forms.
- **Replace text is written as `/StrikeOut`.** The format has no replace-text annotation type; another reader shows the strike and not the substitution, which is visible in the PDF and correct, and incomplete as a representation of the intent.
- **One machine, one engine.** The timings come from a single Windows host and a single pdf.js version.
- **Rendering quality is not evaluated.** The application delegates rasterisation to pdf.js; any rendering difference the reader observes is the engine's, not this project's, and the paper makes no claim either way.

9.3 Conclusion

A PDF reader that writes is a different system from a PDF reader that renders, and the difference is concentrated in a small number of invariants that are easy to state and easy to break: rebuild a page's annotation array rather than appending to it; materialise what a page inherits before flattening its tree; renumber references when copying between documents and leave stream bytes alone. Each invariant, violated, produces a file that opens and is wrong — which is why the verification described here asks questions of pixels and bytes rather than of return values, and why the paper reports the defects that reached a user as well as the ones the suites caught.

The application is 17,323 lines of source over 47 files with one runtime dependency, MIT licensed, and available with its tests and the scripts that rebuilt every figure here at `https://github.com/dongfangxy/pdf-looker-pro`.

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It is in that spirit — and because a program that expects its behaviour to be checked should let its readers check it — that all of this software is released as open source, with nothing held back. The disclosure is deliberate: a reader is entitled to know how a document was produced, as a reader of the source is entitled to see every line of it.

The dual representation of §5, the coordinate convention and the decision to write into the document rather than beside it came from using the software daily while building it. The pdf.js project [2] does the rendering; the writer's invariants were learned by breaking them.

I thank 梁圣人 (Liang Shengren), and this remarkable era, in which such an undertaking is within reach of one person with a laptop and a good interlocutor.

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